



MULTILINGUAL • AI-POWERED • EXAM PREP

# MAT Multilingual AI Tutor

AI-Powered Exam Preparation Assistant for Competitive Exams

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# Problem Statement



## No personalized guidance

Students struggle to find tailored, exam-specific coaching for competitive exams like JEE, CAT and MAT.



## Generic AI lacks focus

General chatbots answer everything and drift off-topic, diluting relevance for serious exam preparation.



## Language barriers

Learners need explanations in their own language, but most tools assume English-only comprehension.



## Weak revision tools

There is no integrated way to revise, practice and self-assess in a single, focused workflow.

WHY IT MATTERS COMMERCIALLY

# Business Use Case



## AI-Powered Assistant

An intelligent educational companion that scales personalized tutoring to thousands of learners at near-zero marginal cost.



## Focused Exam Prep

Purpose-built for competitive exams, increasing outcomes, engagement and student confidence.



## Multilingual Access

Removes language barriers, widening the addressable market across regions and demographics.



## EdTech Opportunity

Slots into student learning platforms, coaching institutes and self-study apps as a differentiated feature.

WHAT WE SET OUT TO BUILD

# Project Objectives

1



## Multilingual learning support

Tutoring in the learner's preferred language.

2



## Focused exam preparation

Every answer anchored to the selected exam & section.

3



## Interactive learning experience

Conversational doubt-solving that adapts to questions.

4



## Strong revision support

Auto-generated notes & flashcards reinforce concepts.

5



## Self-assessment tools

Quizzes with scoring and feedback to track progress.

# Existing System vs Proposed System



## EXISTING SYSTEM

- ✗ Traditional, classroom-style learning methods
- ✗ Generic chatbot tools with no boundaries
- ✗ No exam-specific focus or syllabus awareness
- ✗ No integrated revision or practice tools



## PROPOSED SYSTEM – MAT

- ✓ Exam-focused AI tutor for JEE, CAT, NEET, etc.
- ✓ Multilingual support for wider accessibility
- ✓ Scope-restricted, on-topic responses only
- ✓ Notes, quizzes & flashcards in one platform

# Proposed Solution Overview

MAT guides each learner through a simple, configurable flow that keeps every interaction focused on their chosen exam.

## 1. STUDENT CONFIGURES



Language



Exam



Section



Proficiency

## 2. STUDENT INTERACTS

Asks exam doubts or launches a study tool — notes, quiz or flashcards.



Chat



Notes



Quiz



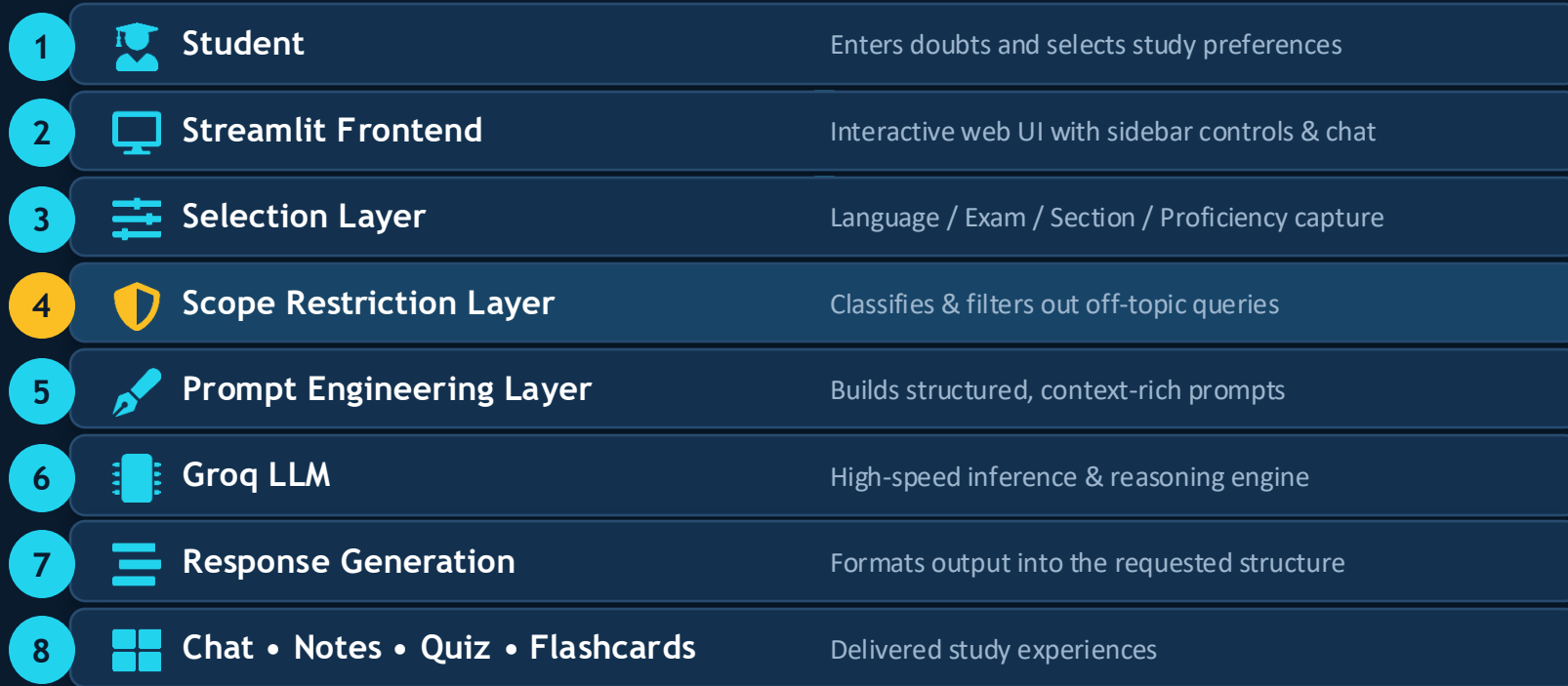
Cards



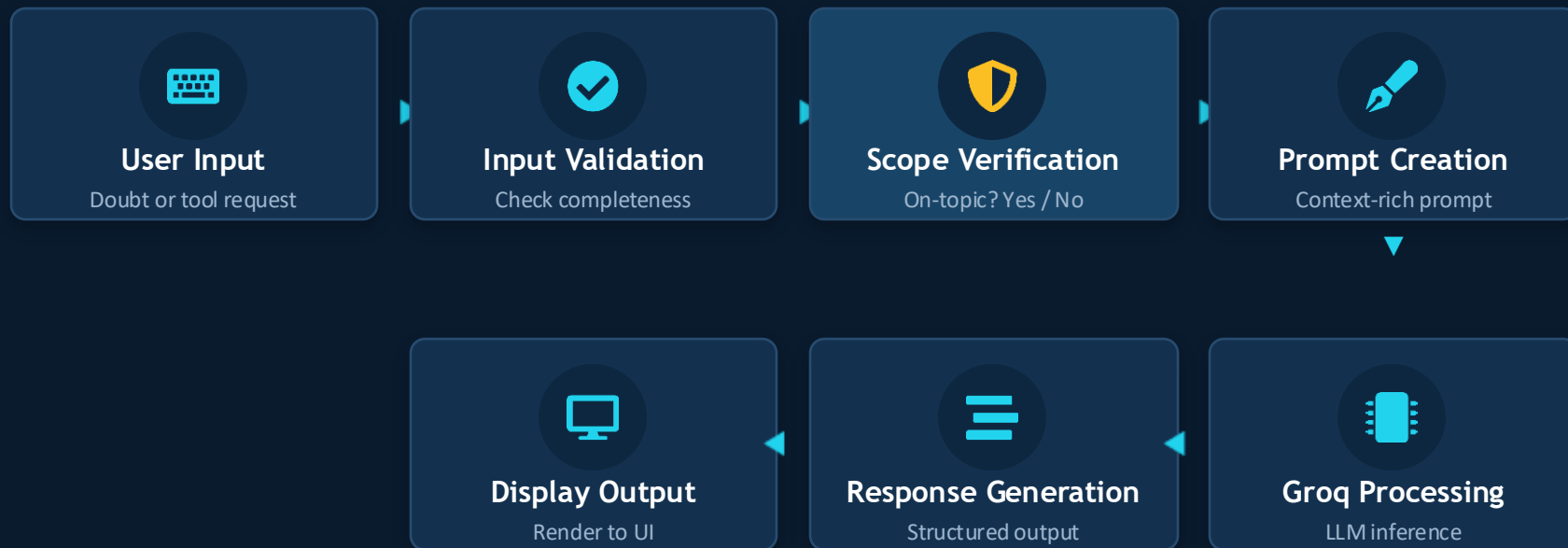
## 3. FOCUSED SUPPORT

The system delivers tailored, exam-relevant help — always in the chosen language, always within scope.

# System Architecture



# Solution Workflow





# Technology Stack



## Python

Core language orchestrating the entire application logic.



## Streamlit

Rapid, interactive web UI with minimal front-end overhead.



## Groq LLM

Ultra-low-latency inference powering tutoring responses.



## Prompt Engineering

Crafts exam, section & language-aware instructions.



## Session State

Holds user preferences during an active session.



## Query Parameters

Persists settings in the URL across page refreshes.



## JSON Outputs

Structured generation for quizzes and flashcards.



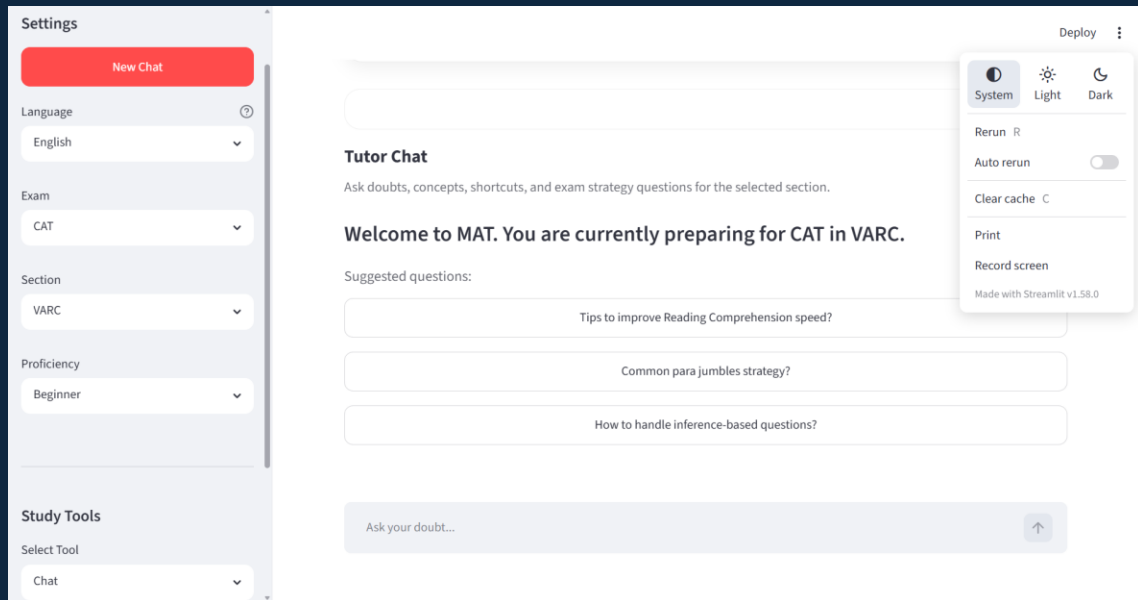
A lightweight, Python-only stack — no heavy framework, fast to build, easy to deploy and maintain.

DESIGNED FOR CLARITY

# User Interface Design

## SIDEBAR & CONTROLS

-  Language selector
-  Exam selector
-  Section selector
-  Proficiency selector
-  Study tools dropdown
-  Clean chat layout
-  Light / Dark mode UI



# Multilingual Support

## SUPPORTED LANGUAGES

English

Hindi

Tamil

Telugu

Bengali

Marathi

Kannada

+ more



### Better understanding

Concepts explained in the student's own language improve comprehension and retention.



### Wider accessibility

Opens quality exam prep to non-English-first learners across regions.



### Higher engagement

Comfort in language builds confidence and keeps students learning longer.

TAILORED TO EACH EXAM

# Exam & Section Based Configuration



**JEE**

*Engineering entrance*

Physics • Chemistry •  
Mathematics



**CAT**

*MBA admissions*

Quant • VARC • DILR



**NEET**

*Medical entrance*

Physics • Chemistry • Biology



**GRE**

*Graduate studies*

Verbal • Quantitative •  
Analytical Writing



**GMAT**

*Business school*

Quant • Verbal • Data Insights



**Section-based learning** targets the exact syllabus area; **proficiency levels (Beginner → Advanced)** adjust depth and pace for personalized preparation.

# Scope Restriction — The Problem



## Why unrestricted LLMs fail exam prep

- LLMs answer ANY question by default — no built-in boundaries.
- Off-topic answers reduce relevance and dilute exam focus.
- Students get distracted and drift away from their syllabus.
- Trust erodes when a 'tutor' behaves like a generic chatbot.

## UNRELATED QUERIES A NAIVE TUTOR WOULD ANSWER



*"Write Python code for a game."*



*"What's the weather today?"*



*"Tell me a joke."*



*"Suggest a movie to watch."*



*"Help me write an email."*

# Scope Restriction – The Solution



## Query Classification

Categorize each incoming question.



## Scope Check

Does it belong to the selected exam & section?



## Refusal Mechanism

Politely decline unrelated queries.



## Redirect

Guide the student back to exam context.

## ✓ ALLOWED

*"Explain projectile motion for JEE Physics."*  
*"Give a CAT quant shortcut for percentages."*

## ✗ REJECTED

*"Write Python code for a game."* → redirected  
*"What's the weather today?"* → redirected

WHERE THE INTELLIGENCE LIVES

# Prompt Engineering Strategy



## Exam-specific

Anchors answers to JEE / CAT / GRE / GMAT / NEET expectations and difficulty.



## Section-specific

Focuses on the exact syllabus area selected by the student.



## Language-specific

Generates responses in the chosen language consistently.



## Proficiency-level

Tunes depth and vocabulary to beginner through advanced.



## Structured prompts

Drive clean JSON for notes, quizzes and flashcards.



Layered prompts combine all four dimensions on every call.

# Chat Module



## Ask doubts naturally

Students type questions in everyday language.



## Answers in-scope only

Responds only to selected-exam questions.



## Greeting handling

Friendly greetings are recognized and welcomed.



## Dynamic output

Exam & section-specific, context-aware replies.

### Tutor Chat

Ask doubts, concepts, shortcuts, and exam strategy questions for the selected section.



Mechanics ke important concepts?



Mechanics ke important concepts samajhne ke liye, humein kuch basic cheezein dhyan mein rakhni hongi.

### Basics of Mechanics

Yeh concept simple hai, basically iska matlab hai ki humein kin kin cheezon ko samajhna hoga:

- **Motion:** Yeh ek aisa concept hai jisme hum dekhte hain ki koi object kitni tezi se ja raha hai, uski displacement kya hai, aur uska acceleration kya hai.
- **Forces:** Forces woh cheezein hain jo kisi object ko move karne ke liye ya uski motion ko badalne ke liye zimmedar hain. Jaise ki friction, gravity, aur normal force.
- **Energy:** Energy ek aisa concept hai jisme hum dekhte hain ki koi object kitna kaam kar sakta hai. Jaise ki kinetic energy, potential energy, aur thermal energy.

Apna doubt pooch...





# Notes Generator Module



Request Notes



Prompt Creation



Groq Response



Structured Notes



Download Option

## MAT - Multilingual AI Tutor

Personalized exam preparation assistant for competitive exams.

Language: English

Exam: JEE Mains

Section: Physics

Level: Beginner

### Quick Notes

Generate concise revision notes for the selected exam section.

JEE Mains Physics Revision Cheat Sheet

### Key Concepts

- Mechanics: Kinematics, Dynamics, Work and Energy, Rotational Motion
- Electromagnetism: Electric Fields, Magnetic Fields, Induction
- Thermodynamics: Laws of Thermodynamics, Thermal Expansion
- Waves and Optics: Wave Motion, Reflection and Refraction

# Quiz Generator Module



## MCQ Generation

Creates exam-style multiple-choice questions on demand.



## JSON Structure

Questions, options & answers returned as clean JSON.



## Answer Evaluation

Compares student responses against the correct keys.



## Score Computation

Calculates and displays the final score instantly.



## Feedback Display

Explains correct answers to reinforce learning.



Practice → evaluate → improve, all in one loop.

# Flashcard Generator Module



## Concept-based generation

Automatically extracts key concepts into flashcards.



## Question - Answer format

Front asks, back reveals — classic study format.



## Active recall & revision

Promotes retrieval practice for durable memory.



FRONT

*"What is Newton's Second Law of Motion?"*



BACK

*"Force equals mass times acceleration ( $F = m \times a$ )."*

# Language Persistence



## THE PROBLEM

Selected language reset to default after every browser refresh.

Students lost their preference and had to re-select repeatedly — a frustrating experience.

## THE SOLUTION



**Session State** Store the selected values in memory.



**Query Parameters** Mirror those values into the URL.



**Sync to URL** Keep URL and selection in step on change.



**Restore on Refresh** Read URL params back after reload.

# Challenges Faced & Solutions

| CHALLENGE                                                                                                               | HOW WE SOLVED IT                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>Scope restriction</b>              |  Added a query-classification gate before the LLM call.   |
|  <b>Language persistence</b>           |  Combined session state with URL query parameters.        |
|  <b>UI inconsistency across pages</b>  |  Centralized styling for a uniform light/dark experience. |
|  <b>Prompt optimization</b>            |  Iteratively refined layered, structured prompts.         |
|  <b>Consistent multilingual output</b> |  Enforced language in every prompt and template.          |

# Model Limitations & Edge Cases



## Hallucinations

LLMs may occasionally produce confident but incorrect facts.



## Ambiguous queries

Vague questions can yield off-target answers.



## Misclassification

Scope check may rarely mislabel a borderline query.



## Prompt dependence

Output quality is tightly coupled to prompt quality.



## Language limitations

Rare terms may translate imperfectly in some languages.



We mitigate these with scope gates, structured prompts & clear disclaimers.

# Guardrails, Safety & Responsible AI



## Off-topic rejection

Unrelated queries are filtered out, not answered.



## Exam-only responses

Stays strictly within the educational domain.



## Controlled behavior

Prompt guardrails constrain how the model responds.



## User safety

No harmful, unsafe or out-of-scope content.



## Educational fairness

Equal, multilingual access for diverse learners.



## Responsible AI

Transparent limits and accountable design choices.

# Demo, Results & Future Enhancements



## DEMO WALKTHROUGH

- Select language
- Select exam
- Ask an exam question
- Scope-restriction example
- Generate notes
- Generate a quiz
- Generate flashcards



## RESULTS

- ✓ Fully functional exam-focused tutor
- ✓ Reliable scope restriction in action
- ✓ Persistent multilingual preferences
- ✓ Notes, quizzes & flashcards working end-to-end
- ✓ A more focused, confident learning experience



## FUTURE

-  Voice support
-  OCR doubt solving
-  Image-based questions
-  Adaptive recommendations
-  Analytics dashboard
-  Mobile app



## CONCLUSION

# A focused, multilingual AI tutor — built responsibly.



### Problem solved

Delivered focused, multilingual, exam-specific tutoring.



### Technical achievement

Streamlit + Groq + prompt engineering, end-to-end.



### Key innovations

Scope restriction & language persistence.



### Overall impact

More accessible, trustworthy and effective exam prep.



### Questions & Answers

Thank you for your time and attention.

MAT — Multilingual AI Tutor